



CIT316 – Advanced Software Design

Practical Week 05: Building with Kanban and Sizing

Preparation:

- Check that your team has all necessary materials (straws, tape).
 - Ensure each team member understands basic Kanban principles and concepts like Work in Progress (WIP) limits.
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Part 1: The Project (Straw Tower Building)

Objective:

Re-construct a straw tower as a team, but this time focusing on applying Kanban methodologies and conducting explicit relative sizing for your tasks. Your goal is to build the tallest, most efficient, and free-standing straw tower that meets specified functional requirements within the time limit.

Group Size: 4-5 students per team.

Materials (per Group):

- 15 drinking straws
- 1 roll of tape

Constraints & Penalties:

- **Time Limit:** You will have 30 minutes in total for the building phase (No "sprints").
- **Free-Standing:** The tower must be free-standing on a flat table surface. It cannot be taped to the table or any other object. *Penalty: -10 points.*
- **Height:** Minimum height of the tower must be 40cm. *Penalty: -1 point per cm under the minimum height.*
- **Weight Test:** The tower must be able to hold two books at its peak for at least 10 seconds. *Penalty: -5 points.*

- **Material Efficiency:** Only the provided materials can be used. Excessive or wasteful use of materials is discouraged. *Penalty: -2 points per straw used beyond 15.*

Bonuses:

- **Tallest Tower:** The team with the tallest successful free-standing tower will receive a bonus. *Bonus: +5 points.*
- **Aesthetic Design:** Creativity and neatness in construction will be acknowledged. *Bonus: +3 points.*
- **Efficient Construction:** Most efficient use of materials and clear work management via the Kanban board. *Bonus: +2 points.*

Part 2: Planning, Sizing, and Kanban Implementation

Step 1: Establish Your Product Backlog & Initial Planning Meeting (10 minutes)

- As a team, quickly define the key deliverables/features for your straw tower based on the requirements (e.g., base foundation, main structure section 1, connecting pieces, peak for weight holder, final construction).
- Create distinct user stories or task cards for each logical deliverable. Write these on sticky notes. For example: "As a team, we need a stable base structure," "We need to construct three straw segments for height."

Step 2: Relative Sizing Session with two point Planning Poker (10 minutes)

- Identify two 'standard' one small task that everyone agrees is simple (e.g., a basic single straw joint) and a complex task (Verifying around 40 joints for its strength). Assign points 2 and 8 respectively.
- Use Planning Poker cards (or a simple alternative like writing numbers 1, 2, 3, 5, 8, 13, 21 on paper) or everyone go online via <https://planningpokeronline.com/> for estimation.
- For each backlog item (sticky note):
 1. Read the task aloud to the team.
 2. Each team member privately selects a number card reflecting their estimate of effort *relative* to the standard tasks. *Think in terms of effort, complexity, and risk, not just time.*
 3. All cards are revealed simultaneously.

4. If estimates differ significantly (e.g., some vote 2, others vote 13), discuss the reasons for the discrepancy. Re-vote as necessary until the team reaches a reasonable consensus on the story points.

Step 3: Setup Your Kanban Board

- Create your Kanban board columns on a whiteboard, wall, or flipchart: **To Do, In Progress, Blocked, Done**. You can add more columns if your team finds them useful (e.g., *Testing/Validation*).
- Define **WIP (Work in Progress) limits** for the "In Progress" column (e.g., no more than 2 items at a time) to prevent bottlenecks and promote continuous flow.
- Populate your **To Do** column with the prioritized sticky note tasks.

Step 4: Execute the Work (The Building Sprints) (30 minutes)

- Run multiple short "building sessions" (e.g., four 6-minute iterations, each separated by a brief 1.5-minute stand-up/review meeting).
- Pull tasks from **To Do** to **In Progress** strictly based on available capacity and your established WIP limits. Focus on finishing current work before starting new tasks.
- Once a task is complete and validated, move it to **Done**. Mark any blockers/roadblocks visually (e.g., with a different color sticky note or a symbol).

Part 3: Final Discussion and Questions

Wrap-Up Discussion (15 minutes)

- Have the groups present their straw towers and successful construction details.
- Lead a discussion comparing this experience with previously used models (like the requirement mapping/waterfall-like structure from Tutorial 1). Ask: "Towards the end, what do you believe is the best model for different types of projects/challenges? What did you find challenging about Kanban? Where did you see benefits?"

